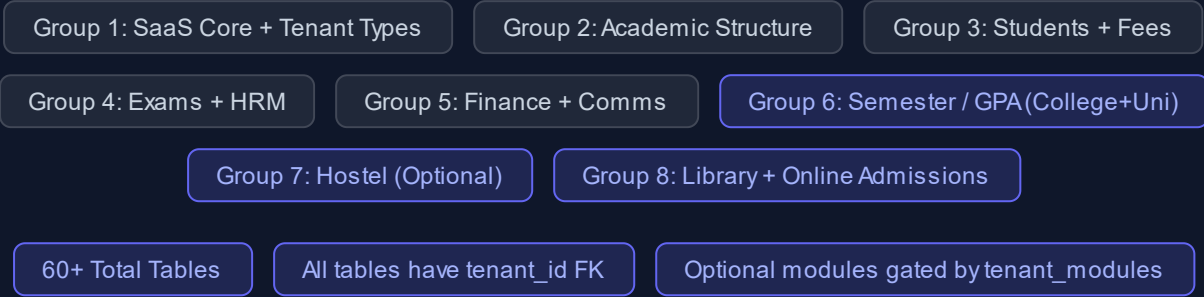


SaaS EMS — Complete ER Diagram

Database Schema — Multi-Institution Laravel SaaS

Version 2.0



tenants SAAS

id	PK bigint
name	string
slug	string unique
domain	string nullable
type	enum school/college/university
logo	string nullable
timezone	string
currency	string default PKR
plan	enum starter/pro/enterprise
is_active	boolean
trial_ends_at	timestamp nullable
created_at / updated_at	timestamps

tenant_modules NEW

id	PK bigint
tenant_id	FK → tenants
module_slug	string e.g. semester_gpa
is_enabled	boolean default false
price_override	decimal nullable
enabled_at	timestamp nullable
enabled_by	FK → users (super admin)
notes	text nullable

subscriptions NEW

id	PK bigint
tenant_id	FK → tenants
plan	enum starter/pro/enterprise
student_limit	integer
base_amount	decimal monthly
addons_amount	decimal add-ons total
total_amount	decimal computed
starts_at / ends_at	date
status	enum active/expired/trial
payment_method	string

users

id	PK
tenant_id	FK nullable → super admin
name, email	string
password	string hashed
role	enum superadmin/admin/staff
is_active	boolean
last_login_at	timestamp nullable

user_permissions

id	PK
user_id	FK → users
module	string e.g. fee
action	enum view/create/edit/delete

user_logs

id	PK
user_id	FK
tenant_id	FK
action	string
model	string e.g. Student
model_id	bigint
ip_address	string
created_at	timestamp

KEY RELATIONSHIPS

tenants 1→∞ tenant_modules (module access control)**tenants** 1→∞ users (staff accounts)**tenants** 1→∞ subscriptions (billing history)**users** 1→∞ user_permissions (granular access)

■ tenant_modules is the key to the whole SaaS — every protected route checks tenant->hasModule('slug') via middleware

■ tenants.type = school/college/university — set at registration, drives UI and module availability

campuses

id	PK
tenant_id	FK → tenants
name, city, address	string
phone	string
is_active	boolean

sessions

id	PK
tenant_id	FK
name	string e.g. 2024-2025
start_date / end_date	date
is_current	boolean

classes

id	PK
tenant_id	FK
name	string e.g. Grade 5, Year 1
order_no	integer

campus_classes

id	PK
campus_id	FK → campuses
class_id	FK → classes
session_id	FK → sessions

sections

id	PK
campus_class_id	FK
name	string e.g. A, B, Blue
capacity	integer

subject_groups

id	PK
tenant_id	FK
name	string e.g. Sciences

subjects

id	PK
tenant_id	FK
subject_group_id	FK
name, code	string
credit_hours	integer nullable (uni)

class_subjects

id	PK
campus_class_id	FK
subject_id	FK
marks_total	integer

shifts

id	PK
tenant_id	FK
name	string e.g. Morning
start_time / end_time	time

grading_policies

id	PK
tenant_id	FK
campus_class_id	FK nullable
grade_label	string e.g. A+
min_percent / max_percent	decimal
grade_points	decimal e.g. 4.0

students

id	PK
tenant_id, campus_id	FK
section_id	FK → sections
family_id	FK → families
reg_no	string unique per tenant
name, father_name	string
cnic, dob, gender	string/date
religion, nationality	string
phone, email, photo	string
status	enum active/inactive/left
house	string nullable (school)

families

id	PK
tenant_id	FK
family_code	string unique
guardian_name, relation	string
phone, email	string

admission_queries

id	PK
tenant_id, campus_id	FK
name, phone	string
class_interest	string
status	enum pending/follow/enrolled
source	string walk-in/online/referral

fee_plans

id	PK
tenant_id	FK
campus_class_id	FK
name	string
is_active	boolean

fee_particulars

id	PK
fee_plan_id	FK → fee_plans
name	e.g. Tuition Fee
amount	decimal
repeat_months	json array
due_day	integer

student_fee_plans

id	PK
student_id	FK
fee_plan_id	FK
session_id	FK
assigned_at	date

challans

id	PK
tenant_id, student_id	FK
session_id	FK
challan_no	string unique
total_amount	decimal
paid_amount	decimal default 0
discount_amount	decimal default 0
fine_amount	decimal default 0
due_date	date
status	enum unpaid/partial/paid

challan_items

id	PK
challan_id	FK
fee_particular_id	FK
name	string (snapshot)
amount	decimal

payments

id	PK
challan_id, student_id	FK
collected_by	FK → users

discount_types

id	PK
tenant_id	FK
name	e.g. Sibling Discount

amount	decimal
method	enum cash/bank/online
transaction_ref	string nullable
paid_at	timestamp

type	enum percent/fixed
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challan_discounts	
id	PK
challan_id	FK
discount_type_id	FK
amount	decimal
reason	text

online_payments NEW	
id	PK
challan_id, tenant_id	FK
gateway	enum jazzcash/easypaisa/bank
gateway_txn_id	string unique
amount	decimal
status	enum pending/success/failed
gateway_response	json
paid_at	timestamp nullable

exams

id	PK
tenant_id, session_id	FK
campus_class_id	FK
name	e.g. Mid-Term 2025
type	enum monthly/mid/final/quiz
exam_date	date

exam_marks

id	PK
exam_id	FK
student_id	FK
subject_id	FK
marks_obtained / marks_total	decimal
grade	string e.g. A+
remarks	text nullable

student_attendance

id	PK
student_id, section_id	FK
date	date
status	enum present/absent/leave/late
remarks	string nullable
marked_by	FK → users

employees

id	PK
tenant_id, campus_id	FK
department_id, designation_id	FK
shift_id	FK
emp_no	string unique
name, cnic, gender	string
phone, email, photo	string
status	enum active/inactive
join_date	date

departments

id	PK
tenant_id	FK
group_id	FK → dept_groups
name	string

designations

id	PK
tenant_id, department_id	FK
name	e.g. Head Teacher

employee_attendance

id	PK
employee_id	FK
date	date
time_in / time_out	time nullable
status	enum present/absent/leave
method	enum manual/qr/biometric

salary_plans

id	PK
employee_id	FK
basic_salary	decimal
frequency	enum monthly/weekly
effective_from	date
is_active	boolean

salary_allowances

id	PK
salary_plan_id	FK
name	e.g. House Rent
amount	decimal
type	enum allowance/deduction

salary_payments

id	PK
employee_id, salary_plan_id	FK
month, year	integer
gross / deductions / net	decimal
status	enum draft/paid
paid_at	timestamp

leave_requests

id	PK
employee_id	FK

sms_logs

id	PK
tenant_id	FK

type	enum sick/casual/annual
from_date / to_date	date
reason	text
status	enum pending/approved/rejected
approved_by	FK → users

phone, message	string
type	enum fee/attendance/result/blast
status	enum sent/failed/pending
gateway	string
sent_at	timestamp

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Finance (GL/Accounting) & Communication

Double-entry GL, journal entries, financial statements, WhatsApp, notifications, AI logs

8 tables

gl_groups

id	PK
tenant_id	FK
name	e.g. Current Assets
type	enum asset/liability/equity/income/expense

accounts

id	PK
tenant_id, gl_group_id	FK
code	string e.g. 1001
name	string e.g. Cash in Hand
type	enum debit/credit normal
is_active	boolean

journal_entries

id	PK
tenant_id	FK
ref_no	string unique
entry_date	date
description	text
status	enum draft/posted/voided
created_by	FK → users

journal_lines

id	PK
journal_id	FK
account_id	FK → accounts
debit	decimal default 0
credit	decimal default 0
narration	string

whatsapp_logs **NEW**

id	PK
tenant_id	FK
recipient	string phone
template	string template name
message	text
status	enum sent/delivered/failed
sent_at	timestamp

notifications

id	PK
tenant_id, user_id	FK
title, body	string
type	string fee/attendance/system
is_read	boolean default false

ai_report_logs **NEW**

id	PK
tenant_id, user_id	FK
report_type	string e.g. attendance_insight
prompt	text
result	text AI response
tokens_used	integer
generated_at	timestamp

programs **NEW**

id	PK
tenant_id, campus_id	FK
name	e.g. BS Computer Science
code	string e.g. BSCS
degree_level	enum bachelors/masters/phd
duration_years	integer e.g. 4
total_credit_hours	integer e.g. 130
department_id	FK

semesters **NEW**

id	PK
tenant_id, session_id	FK
name	e.g. Fall 2024
semester_no	integer e.g. 1, 2, 3
start_date / end_date	date
is_active	boolean

semester_courses **NEW**

id	PK
semester_id	FK → semesters
subject_id	FK → subjects
program_id	FK → programs
instructor_id	FK → employees
max_seats	integer

course_enrollments **NEW**

id	PK
student_id	FK → students
semester_course_id	FK
status	enum enrolled/dropped/completed
enrolled_at	timestamp

course_grades **NEW**

id	PK
enrollment_id	FK → course_enrollments
marks_obtained	decimal
letter_grade	string e.g. A, B+
grade_points	decimal e.g. 4.0
entered_by	FK → users

gpa_records **NEW**

id	PK
student_id	FK → students
semester_id	FK → semesters
credit_hours_attempted	integer
credit_hours_earned	integer
semester_gpa	decimal e.g. 3.75
cumulative_gpa	decimal CGPA

☐ All tables in this group are only accessible if tenant_modules has semester_gpa enabled for that tenant

hostel_blocks NEW

id	PK
tenant_id, campus_id	FK
name	e.g. Boys BlockA
gender	enum male/female/mixed
warden_id	FK → employees
total_capacity	integer

hostel_rooms NEW

id	PK
block_id	FK → hostel_blocks
room_no	string
floor	integer
type	enum single/double/triple
bed_count	integer
facilities	json e.g. ["AC", "WiFi"]
status	enum available/full/maintenance

hostel_allocations NEW

id	PK
student_id	FK → students
room_id	FK → hostel_rooms
bed_no	integer
allocated_at	date
vacated_at	date nullable
status	enum active/vacated

hostel_fee_plans NEW

id	PK
tenant_id	FK
name	e.g. Double Room Monthly
room_type	enum single/double/triple
amount	decimal
frequency	enum monthly/semester

☐ Hostel challans reuse the main challans table with a hostel_type flag — no separate payment system needed

library_books NEW

id	PK
tenant_id	FK
title	string
author	string
isbn	string nullable
publisher, edition	string
category_id	FK → library_categories
total_copies	integer
available_copies	integer computed

library_categories NEW

id	PK
tenant_id	FK
name	e.g. Science, Reference

library_issues NEW

id	PK
book_id	FK → library_books
member_id	FK → library_members
issued_at	date
due_date	date
returned_at	date nullable
fine_amount	decimal default 0
status	enum issued/returned/overdue

library_members NEW

id	PK
tenant_id	FK
member_type	enum student/employee
member_id	bigint polymorphic
max_books	integer e.g. 3
is_active	boolean

online_admissions NEW

id	PK
tenant_id, campus_id	FK
program_id	FK → programs
tracking_code	string unique public
applicant_name	string
father_name	string
cnic, dob, gender	string/date
phone, email	string
qualification_marks	decimal
documents	json file paths
status	enum pending/review/accepted/rejected
app_fee_paid	boolean
converted_student_id	FK → students nullable
reviewed_by	FK → users nullable

admission_sessions NEW

id	PK
tenant_id	FK
name	e.g. Fall 2025 Admissions
start_date / end_date	date
is_open	boolean
total_seats	integer

Complete Table Summary — 62 Tables Total

Group 1 — SaaS Foundation

tenants, tenant_modules, subscriptions,
users, user_permissions, user_logs

Group 2 — Academic Core

campuses, sessions, classes,
campus_classes, sections,
subject_groups, subjects, class_subjects,
shifts, grading_policies

Group 3 — Students & Fees

students, families, admission_queries,
fee_plans, fee_particulars,
student_fee_plans, challans,
challan_items, payments, discount_types,
challan_discounts, online_payments

Group 4 — Exams & HRM

exams, exam_marks, student_attendance,
employees, departments, designations,
employee_attendance, salary_plans,
salary_allowances, salary_payments,
leave_requests, sms_logs

Group 5 — Finance & Comms

gl_groups, accounts, journal_entries,
journal_lines, whatsapp_logs,
notifications, ai_report_logs

Group 6 — Semester/GPA (College+)

programs, semesters, semester_courses,
course_enrollments, course_grades,
gpa_records

Group 7 — Hostel (Optional)

hostel_blocks, hostel_rooms,
hostel_allocations, hostel_fee_plans

Group 8 — Library + Admissions

library_books, library_categories,
library_issues, library_members,
online_admissions, admission_sessions